

Jeongin Kim

M.S. student in software testing. LLM-driven generation of test environments, and predictive modeling on industrial process data.

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Education

2025.03–present	M.S. Student, Computer Science and Engineering <i>Kyungpook National University</i> Advisor: Prof. Woojin Lee, Software Testing Lab (STLAB). Thesis topic not yet fixed.	Daegu, Korea
2023.03–2024.02	B.Eng., Computer Information Engineering <i>Yeungjin University (Advanced Major Course)</i>	GPA 3.50/4.50 Daegu, Korea
2018.03–2023.02	Associate Degree in Engineering, Web Database <i>Computer Information Division, Yeungjin University</i>	GPA 3.05/4.50 Daegu, Korea
2018.10–2020.06	Republic of Korea Army, mandatory military service (completed)	

Research Interests

Software testing, LLM-based test generation, SIL verification environments, predictive modeling on industrial process data

Publications

*corresponding author, own name in bold

Domestic Conference Papers

all in Korean

- [1] **Jeongin Kim**, Deokyeop Kim, Hyeryung Suh, and Woojin Lee*. Automatic Generation of Test Environment Behaviors Using LLM for Embedded Software Testing. *Proc. Korea Computer Congress (KCC)*, pp. 582–584, June 2026. Code: github.com/99JIK/KCC2026
- [2] **Jeongin Kim** and Woojin Lee*. A Study on the Utilization of Sampling Time-point Data for Performance Enhancement of the Biodegradable Fiber Spinning Process. *Symposium on Ubiquitous Computing and Web Information Technology (UCWIT)*, KIISE Yeongnam Branch, November 2025.
- [3] **Jeongin Kim** and Woojin Lee*. The Ability of Generative AI to Understand and Utilize Test Cases. *Proc. Korea Computer Congress (KCC)*, pp. 397–399, July 2025.

Patents

- [1] **Jeongin Kim**, Deokyeop Kim, and Woojin Lee. A Method for Selecting Robust Optimal Process Parameter Combinations. Provisional filed, full application in progress (July 2026). Applicant: KNU Industry-Academic Cooperation Foundation. MOTIE/KIAT P0022335.

Experience

2024.12–present	Graduate Researcher <i>Software Testing Lab, Kyungpook National University</i>	Daegu, Korea
	• LLM-driven generation of environment object behaviors for SIL testing. A three-stage decomposition-induction-synthesis prompt pipeline more than doubled coverage over zero-shot and few-shot baselines across 11 objects in three domains (elevator, drone, smart factory), consistent on GPT-4 and Llama-3 70B (KCC 2026) • Sampling time-point introduced as a latent variable for temporal batch effects in tensile-strength prediction for biodegradable fiber spinning. With GBR, R^2 rose from 0.724 to 0.811, RMSE fell 17.2%, and accuracy within the industry tolerance rose from 87.1% to 92.4%. MOTIE/KIAT project P0022335 (UCWIT 2025) • Methodology for building software-in-the-loop verification environments at lower cost (2026, ongoing)	
2022.09–2024.06	Researcher (reverse engineering) <i>Arkdata</i>	Daegu, Korea
	• Reverse-engineered transaction log formats across heterogeneous DBMSs (Oracle, PostgreSQL, MariaDB/MySQL, Tiberio) to support development of a Change Data Capture (CDC) solution • Designed and built a Kafka-based real-time pipeline across heterogeneous databases • Implementation, analysis, and testing on a three-person research team	

Teaching

2025.03–present	Teaching Assistant <i>School of Computer Science and Engineering, Kyungpook National University</i>	Daegu, Korea
	• Introduction to Programming: Spring 2025, Summer 2025, Winter 2025–26, Summer 2026 (four terms) • System Programming and Creative Convergence Design: Fall 2025 • Java Programming: Spring 2026	

Projects

2025–present	ICT-ABB Convergence Major Project <i>Kyungpook National University</i>	github.com/99JIK/ABB
	Real-time emotion classification from facial action units. Up to 30 faces at once, IoU-based multi-face tracking, event triggers on emotion changes and AU thresholds, FastAPI REST and WebSocket streaming. MediaPipe, TensorFlow.	
2026.01–2026.03	ST Youngwon Smart Office <i>Contract work, in production use</i>	github.com/99JIK/ST_YOUNGWON
	• RAG question answering over internal documents. Regulatory documents chunked hierarchically (article, paragraph, fixed size), retrieved by hybrid keyword and vector search • LLM providers (OpenAI, Claude, Gemini, local Ollama) behind a protocol abstraction so they can be swapped, per the client's maintainability requirement • FastAPI, ChromaDB, SQLite, Synology NAS (FileStation API) integration, JWT auth, Docker Compose deployment	
2025.09–2025.12	FOCUS (Facial Observation & Concentration Understanding System) github.com/99JIK/FOCUS_KNU_2025_COMP0738 <i>Team coursework project</i>	
	Measures a learner's concentration from webcam video in real time. Scores distraction from five signals: eye closure, head orientation, gaze with head-rotation compensation, yawning, and facial action units. MediaPipe Face Landmarker, vanilla JS, Vite.	

Personal project, Python

Compares abstract syntax trees to judge similarity between submissions, ignoring variable names and comments. Supports Python (built-in `ast`), C/C++ (`clang`), and Java (`javalang`). Adjustable similarity threshold, side-by-side diff viewer. Built for the programming exams I assisted.

Technical Skills

Languages	Python, Java, C/C++, TypeScript, JavaScript
Web / Backend	FastAPI, Spring, Node.js, React, Vue.js
Data / ML	Oracle, PostgreSQL, MySQL, ChromaDB, scikit-learn, LightGBM
Tools	Docker, Git, Linux, Kafka